4/28/25

1.5V LDO Design Project

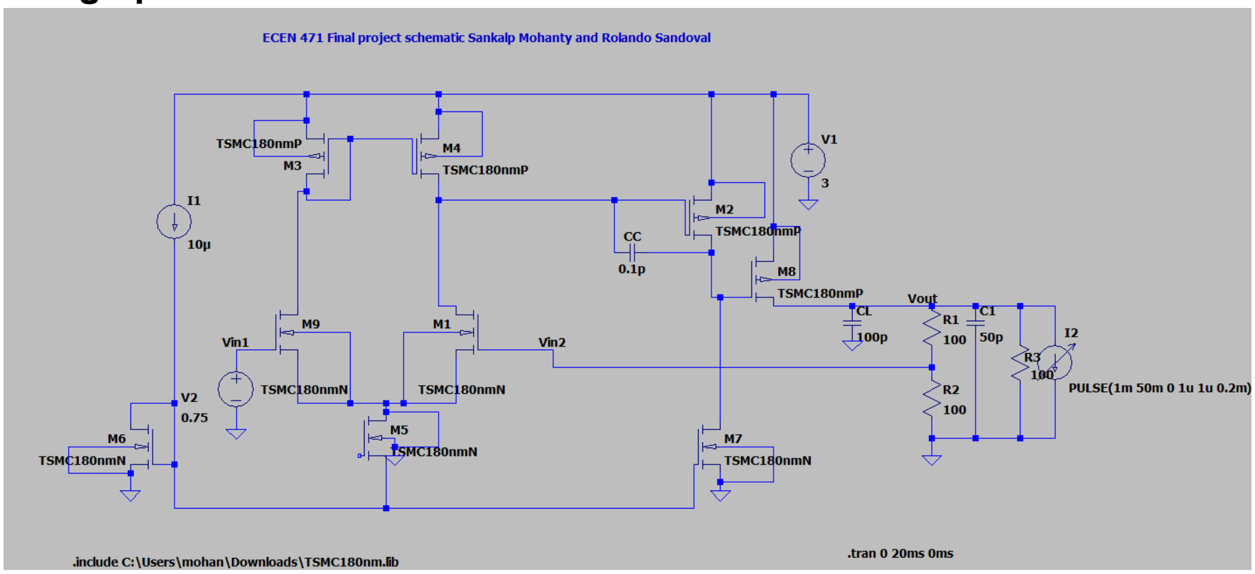
Team members:

Sankalp Mohanty & Rolando Sandoval

Justification and description of chosen architecture:

The 1.5V LDO was the chosen project. We chose this project to gain a deep understanding on how this circuit regulates the output voltage when there is a very low potential difference between the input and output. Our architecture consists of a error amplifier which is a two stage op amp(first stage is a differential pair and the second stage is a gain stage with miller compensation), a pass transistor, and a feedback network to set the reference voltage.

Design procedure:



schematic of project

The 2 V rail on the right produces the current-source bias on the left I1 = 10 µA, this known current sets other biases in the amplifier. The PMOS mirror pair M3–M4, creates currents inside the error-amplifier core. The NMOS M6 is tied to a 0.75 V reference source and the tail device M5 form a 1 µA differential pair which consists of M9 NMOS which is negative feedback and M1 which is positive feedback. When the resistor divider output voltage at Vin2 drifts from 0.75 V, the current split in the pair becomes very unbalanced, creating a voltage swing that transistor M2 amplifies. That gain stage, controlled by transistor M2, is Miller-compensated with a 0.1 pF capacitor to set a dominant pole and inject a phase boosting zero to stabilize the loop gain as we learned in previous labs. The PMOS pass transistor M8 which a larger width relative to the other transistors, drives current to Vout so the feedback divider of R1 = R2 (both 100 Ω) holds Vout at twice the reference of 0.75V, i.e., 1.5 V. A small lead capacitor of 50 pF across the output adds an extra zero that improves the phase margin by adding a phase boost. We can monitor Vout with the transient simulation at the 100 pF output capacitor with a decreased magnitude but increased settling time. Finally, a pulsed current load I2 swings from 1 mA to 50 mA with to verify that the loop limits overshoot or swing to under 100 mV while the amplifier itself draws only about ~500uW of quiescent power, meeting the specification.

Tables of relevant parts:

Transistor	width	length
M1	1.8u	0.35u
M2	220u	0.35u
M3	0.9u	0.35u
M4	0.9u	0.35u
M5	1u	9u
M6	2u	0.5u
M7	150u	9u
M8	330u	0.35u
M9	1.8u	0.35u

Table 1. Transistor sizes

Component	Value
Bias current	10uA
Resistor Dividers(R1 & R2)	100ohms
Miller capacitor	0.1pF
load Caps	100pF and 50pF
Pulse current at load	1mA to 50mA, rising and falling time of 1uS.
Rout	100ohms

Table 2. Component tables

Power consumption of the error amplifier when operating in standby:

$$3V * (10uA + 150uA + 1uA) = 483uW$$

This calculation was made from the input source of 3V and the biasing current sources from transistors M6, M5, and M7.

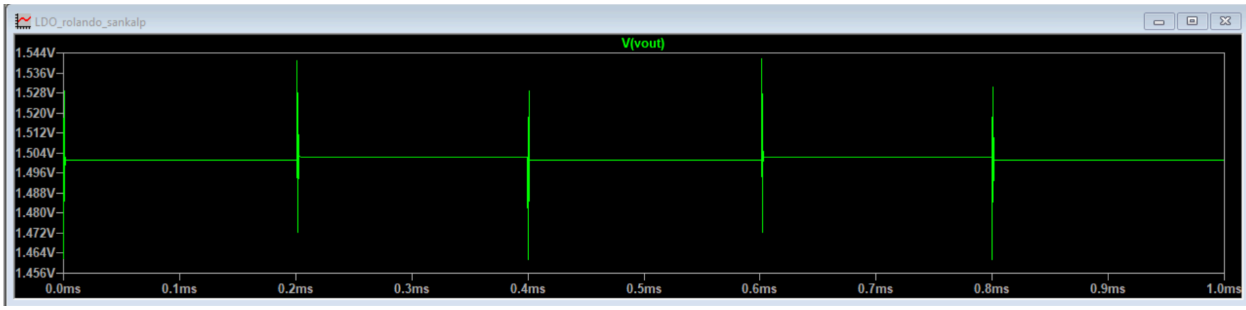


Figure 1: Output Voltage Waveform

The max voltage swing is is **38.7mV** as the lowest the voltage spikes down per a pulse is 1.4613V. So this meets specification.

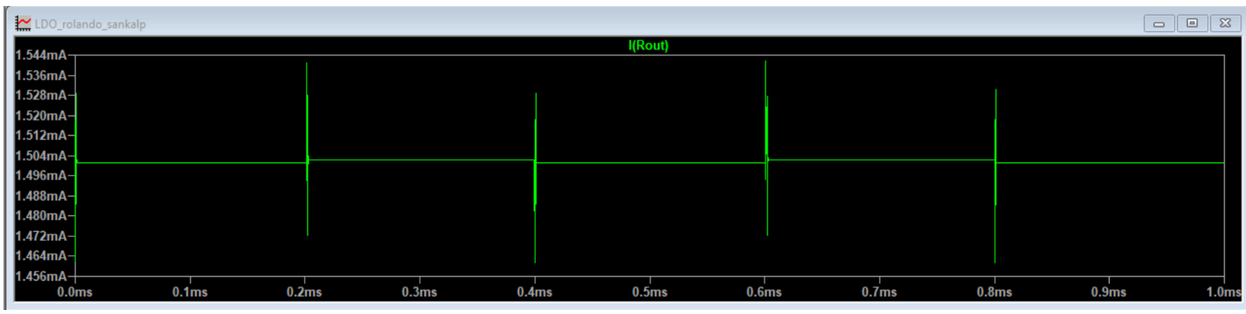


Figure 2: Output Current Waveform(Max load current is under 50mA)

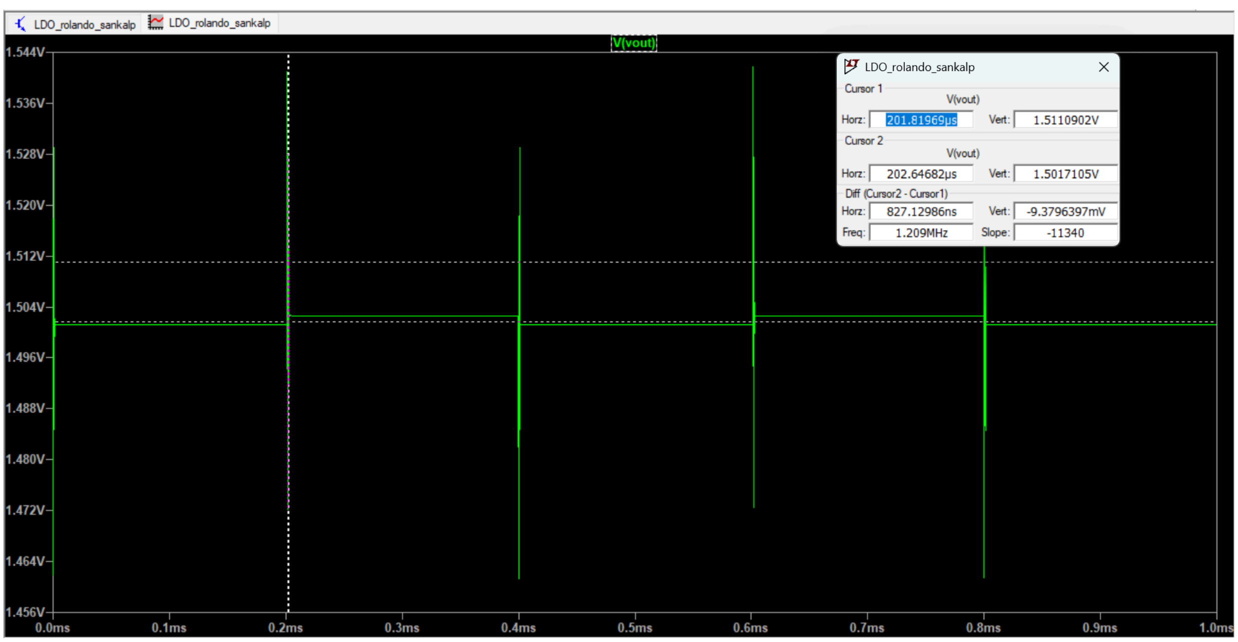


Figure 3: Output Voltage Waveform with Vmin and Vmax

Load regulation = $\frac{Vmin - Vmax}{Vnominal} = \frac{1.511 - 1.501}{1.506} * 100 = 0.664\%$

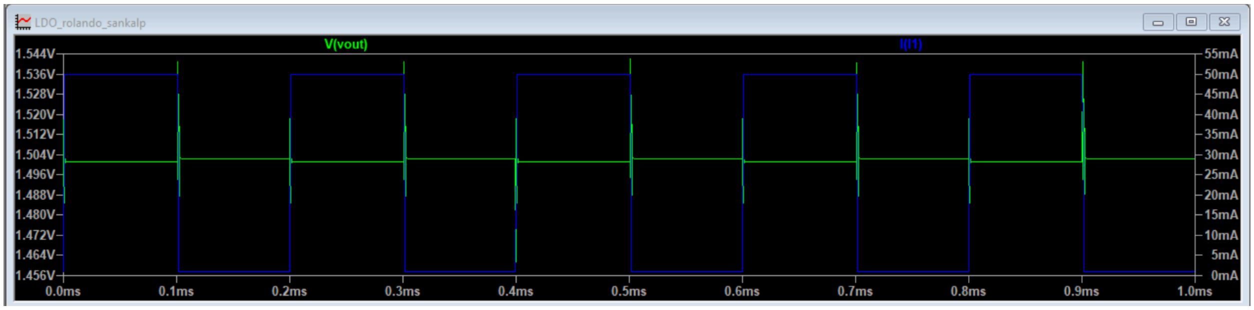


Figure 4: Output Voltage Waveform with Pulsed Current from 1mA to 5mA

Conclusions:

In conclusion we were able to meet all the specifications as outlined. A lot of parameters such as reference voltages, transistor sizes, and zero placement had to be tweaked to ensure we meet design requirements and a sufficient loop gain. This was a very important project to learn how different parts of an LDO including the error amplifier, pass devices, and compensations work together to produce an output to meet our specifications.